

Lab 9

Name: B SHIVA KUMAR

Enrollment No: 92301733056

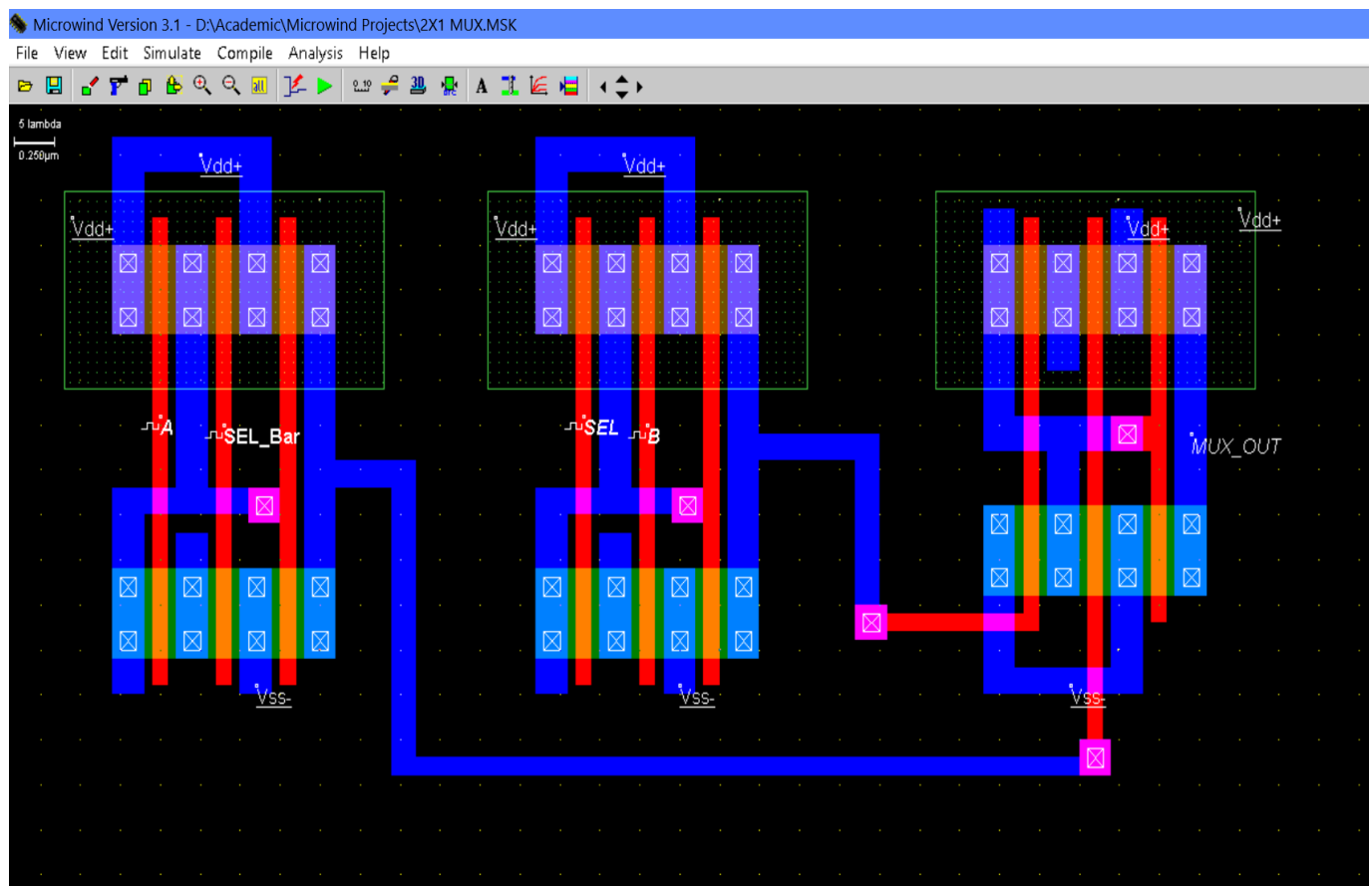
CO1: Develop fundamentals in fabrication of MOSFETs and MOS transistor

CO2: Analyze MOS inverters with its static and switching characteristics

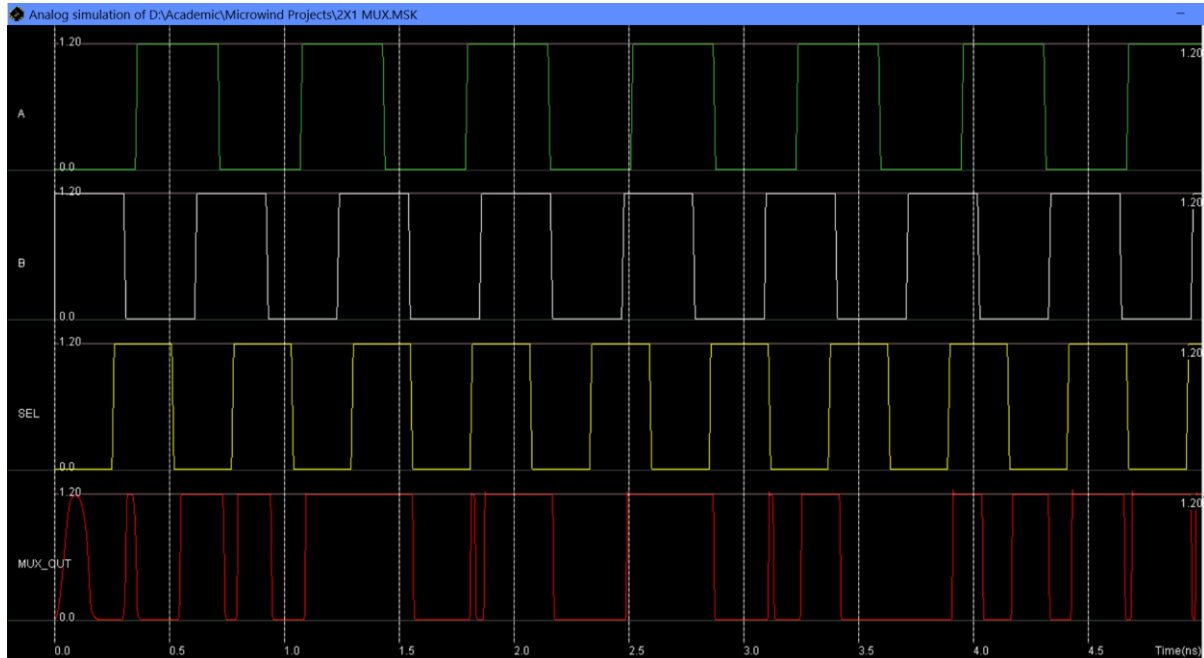
CO3: Analyze Combinational and Sequential MOS logic circuit

Experiment 9: To implement the 2x1 MUX and 4x1 MUX using Transmission gate and simulate the design

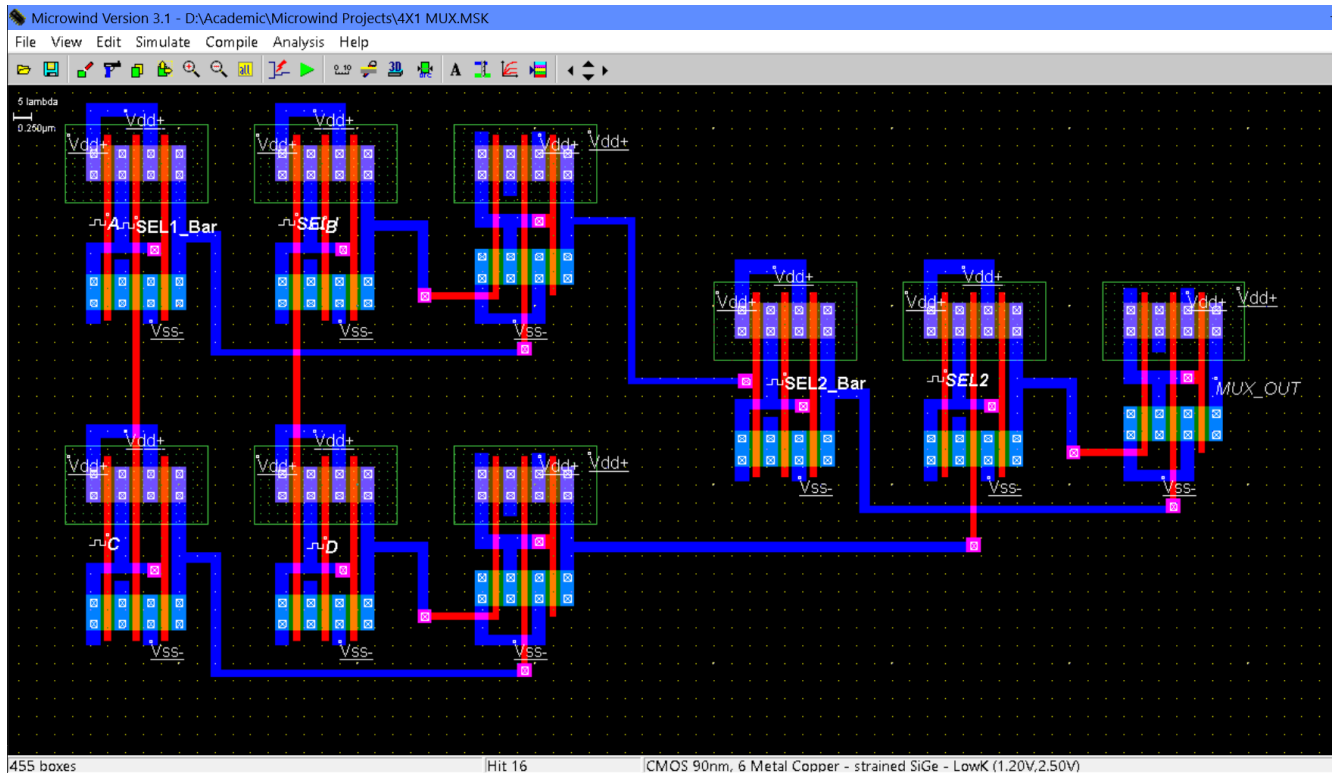
2X1 MUX Layout:



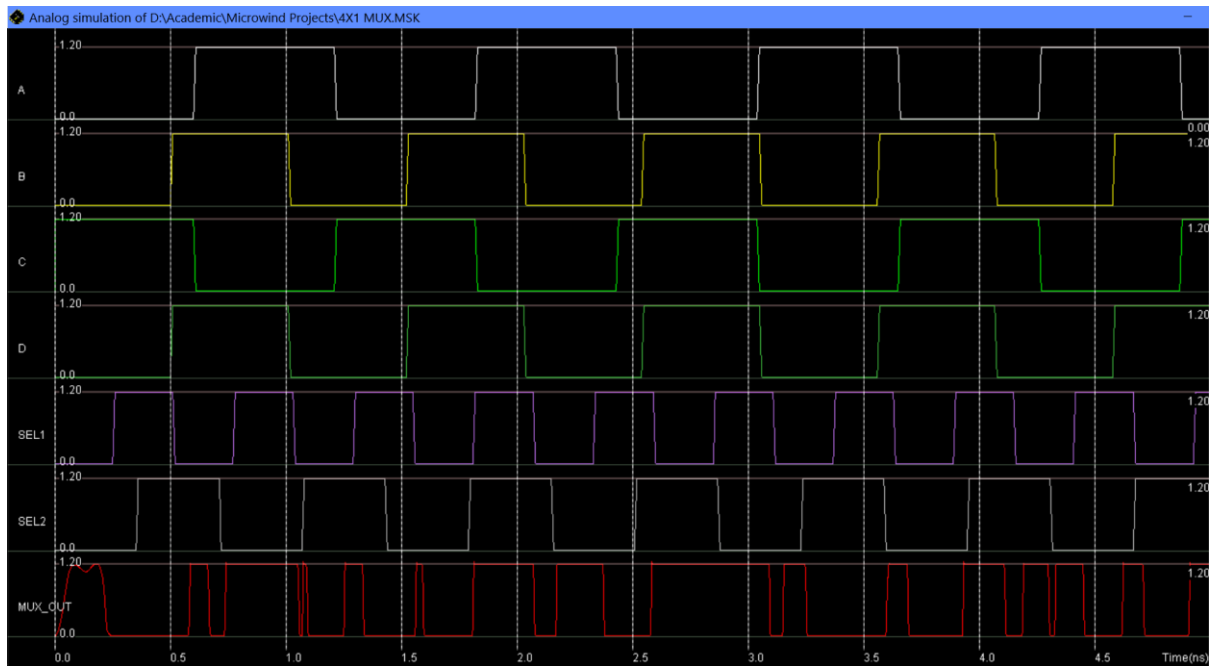
Output Waveform:



4X1 MUX Layout:



Output Waveforms:



Conclusion:

In this experiment, I designed the layout of a 2×1 multiplexer in the Microwind tool using default NMOS and PMOS transistors. By using this 2×1 multiplexer as a basic building block, a 4×1 multiplexer was successfully designed. I also observed the related waveforms and verified them with the truth table.